

محکمہ سائنس و
تکنیک

ML Models → Training

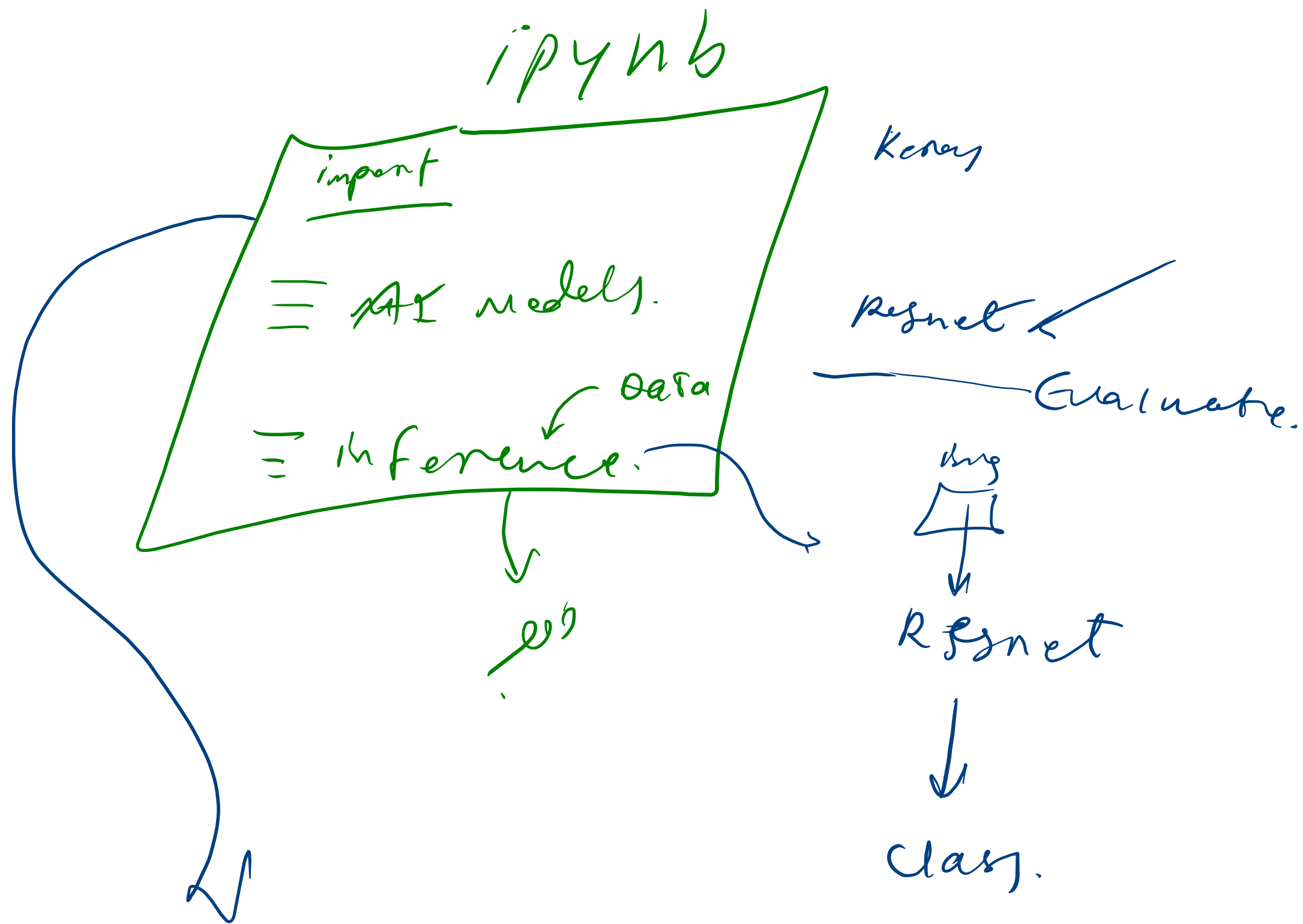
↓
C, we

↓
ML Ops

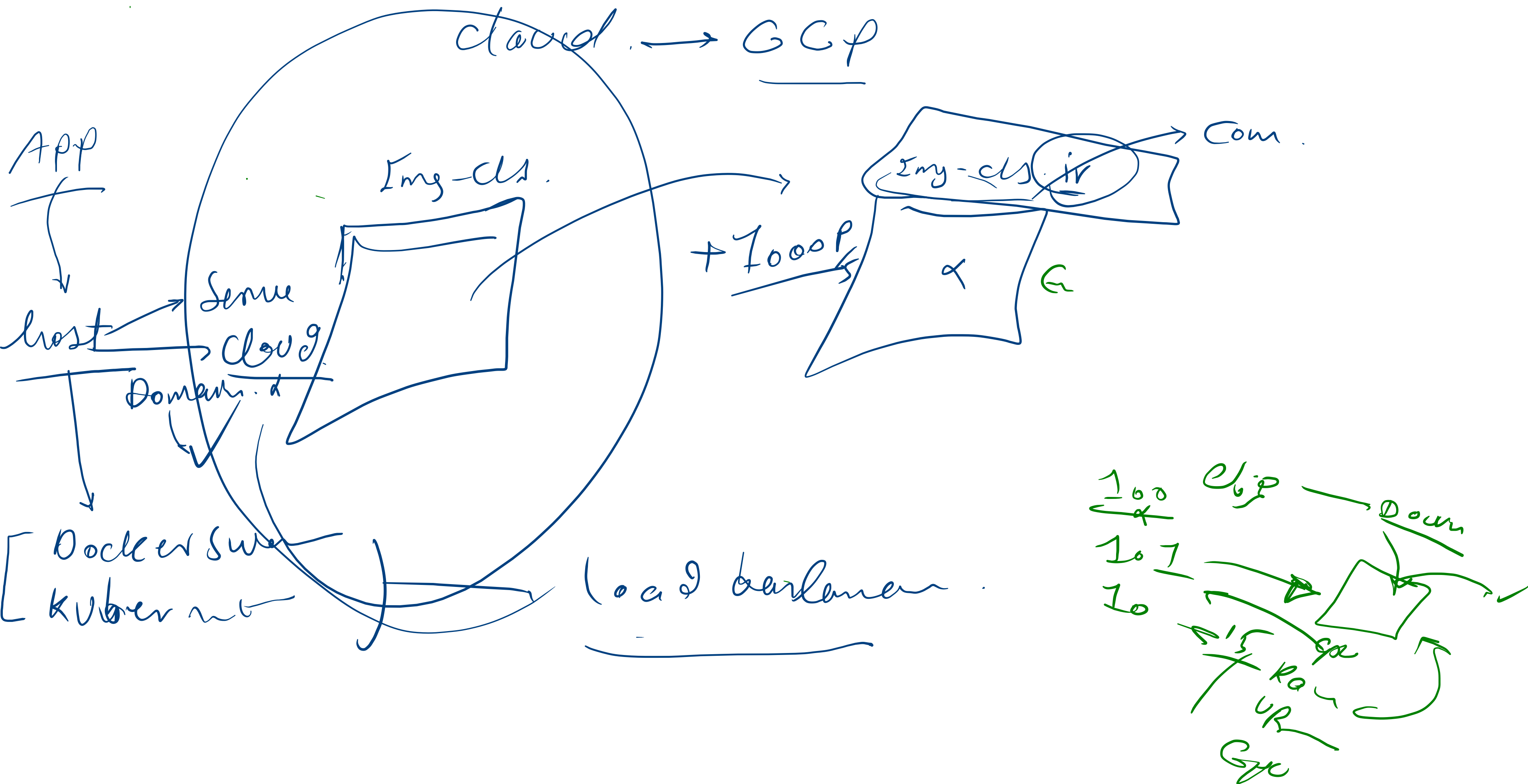
↓
ML Ops

ML Operations

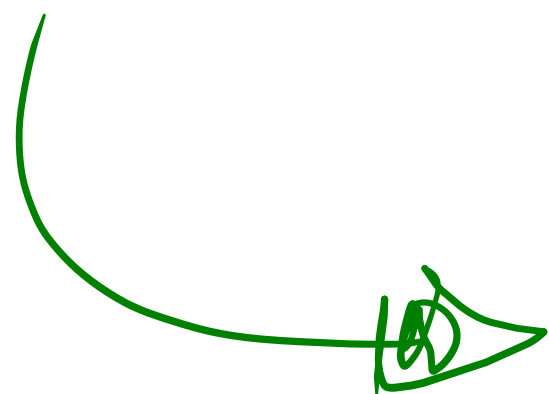
ML Ops



No



End User.



B2C

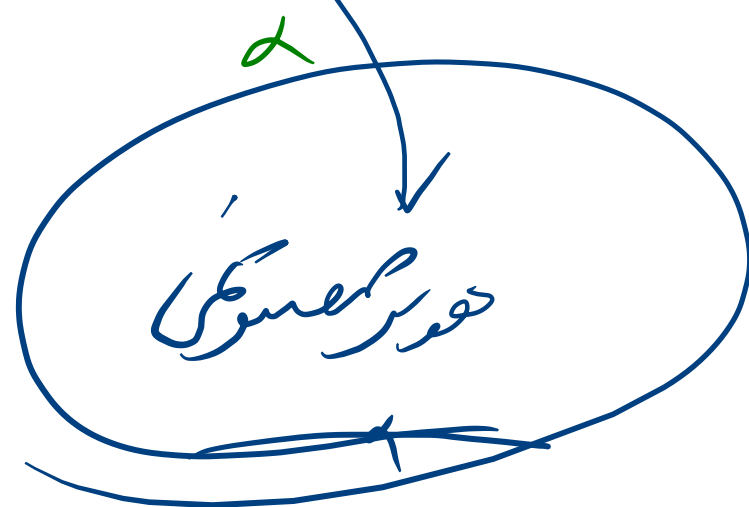
B2B ✓

B2G

رشته‌فروشند عتبی هست برو



رشته
۱
۲



Notes

What ?

Why ?

How ?

✓

What ?

آیا می‌توانیم به سوار شتری در سید قمیته جان پاسخ دهیم؟

CRISP-DM

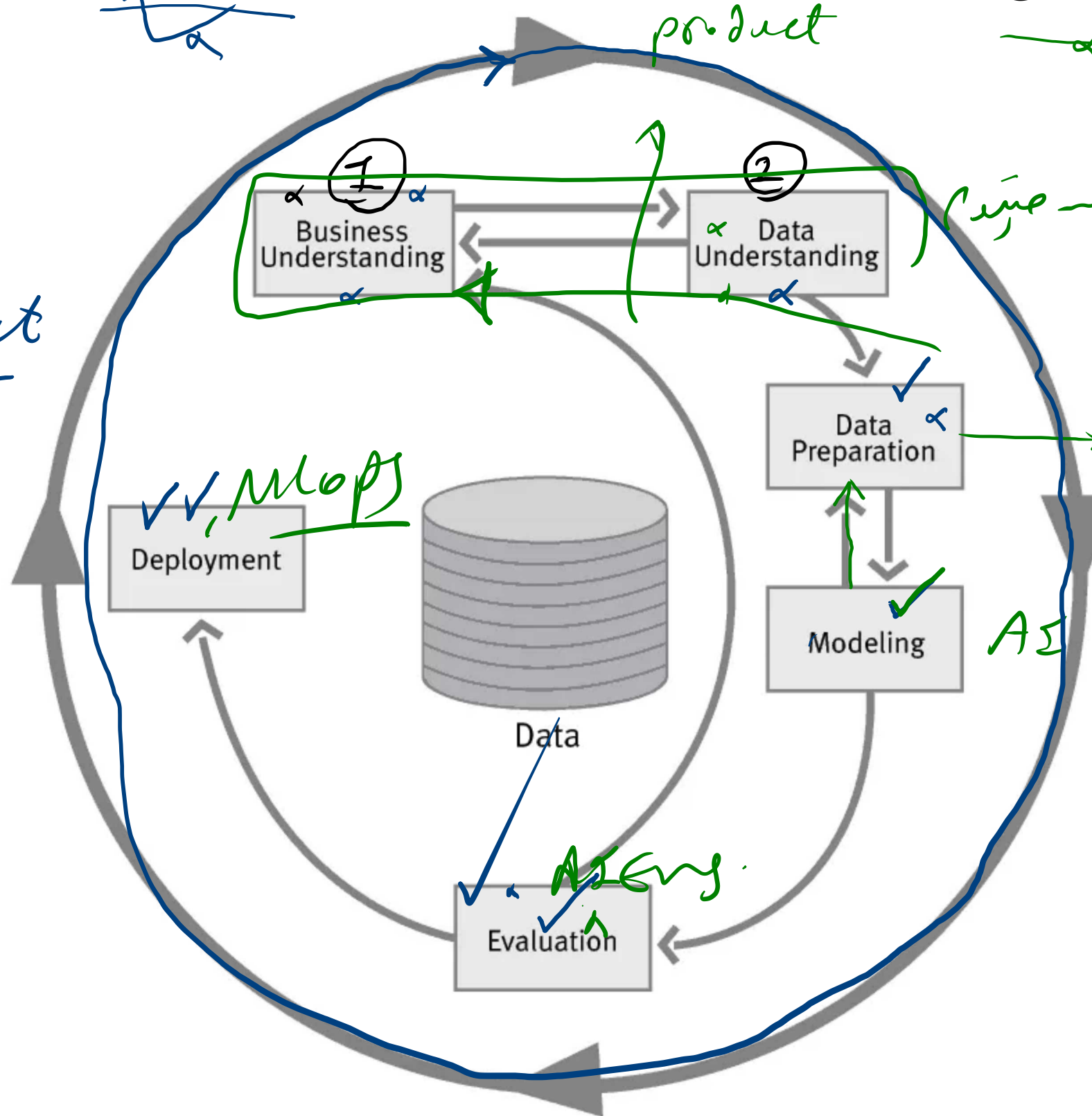
Cross-Industry Standard Process for Data Mining.

مکانیسم / ... / سیستم

نظیر نیکی رفعتی در سید
 چت با داده : chat with data.

AI
 product management

100 ✓
 101
 102
 ...



نیکی ضیاء آن در صفحه اینستاگرام
 CSV

Query

α	γ	γ	α

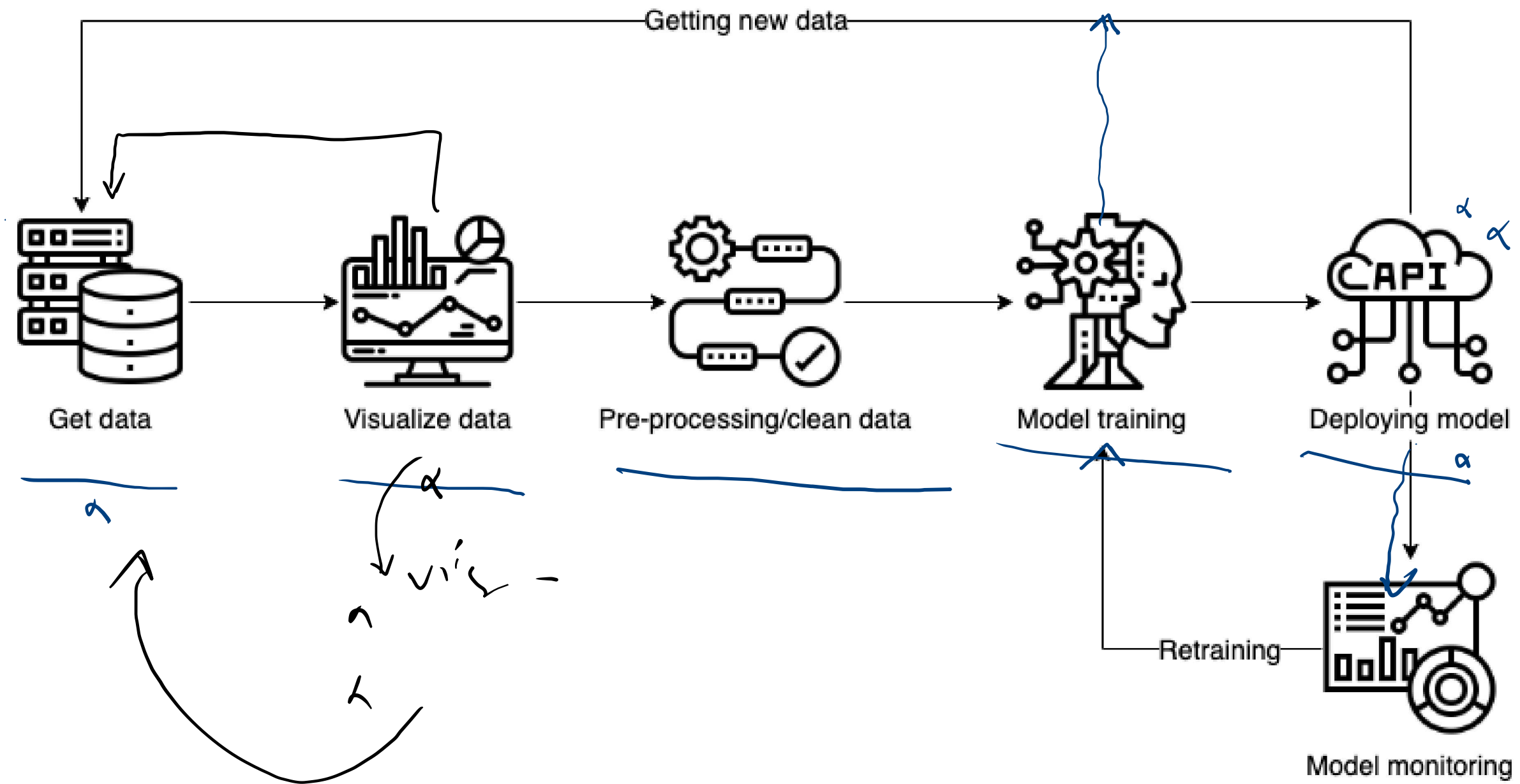
AI
 DE

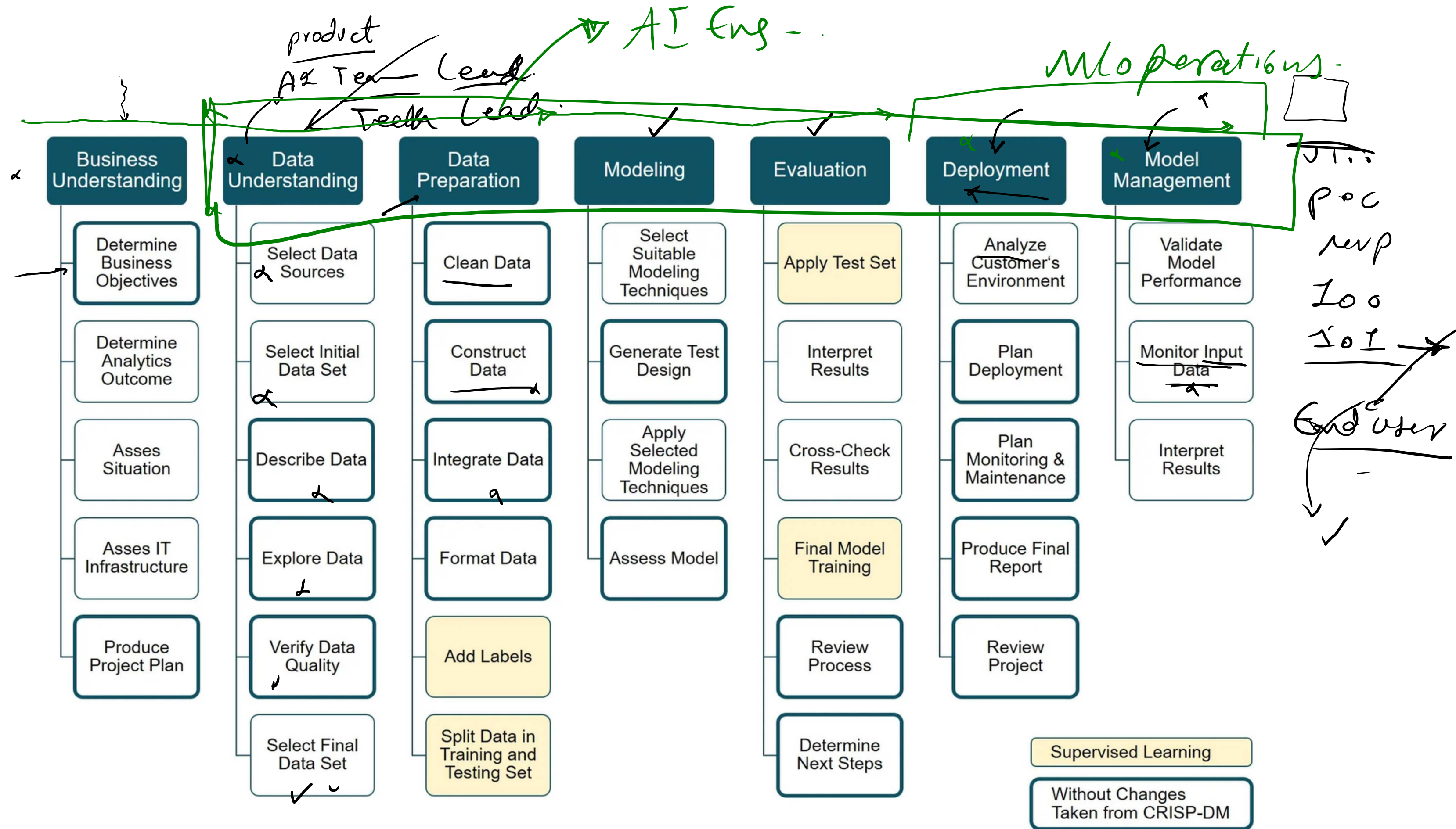
AI Eng.

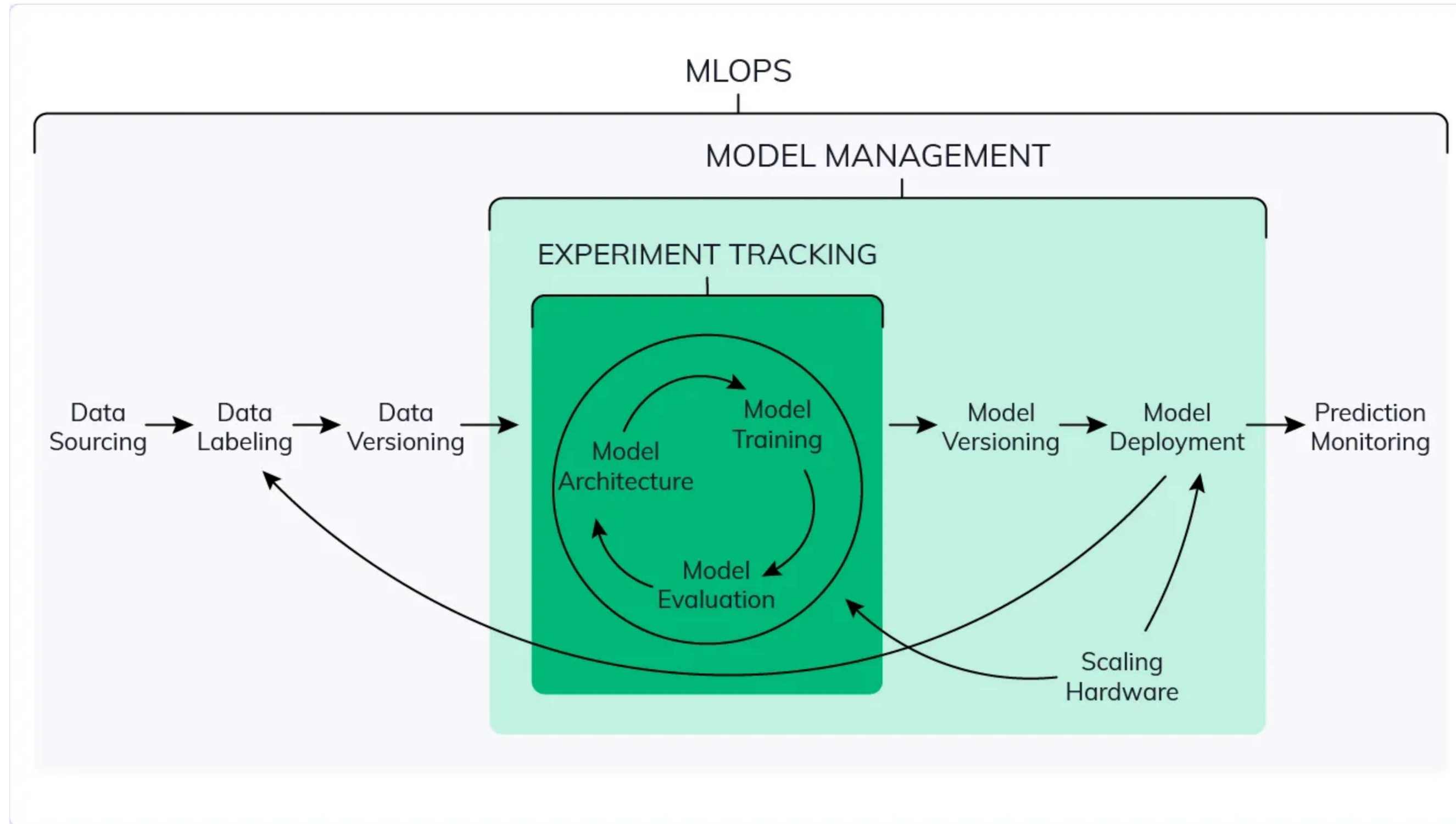
AI Eng.

✓✓ Mlops

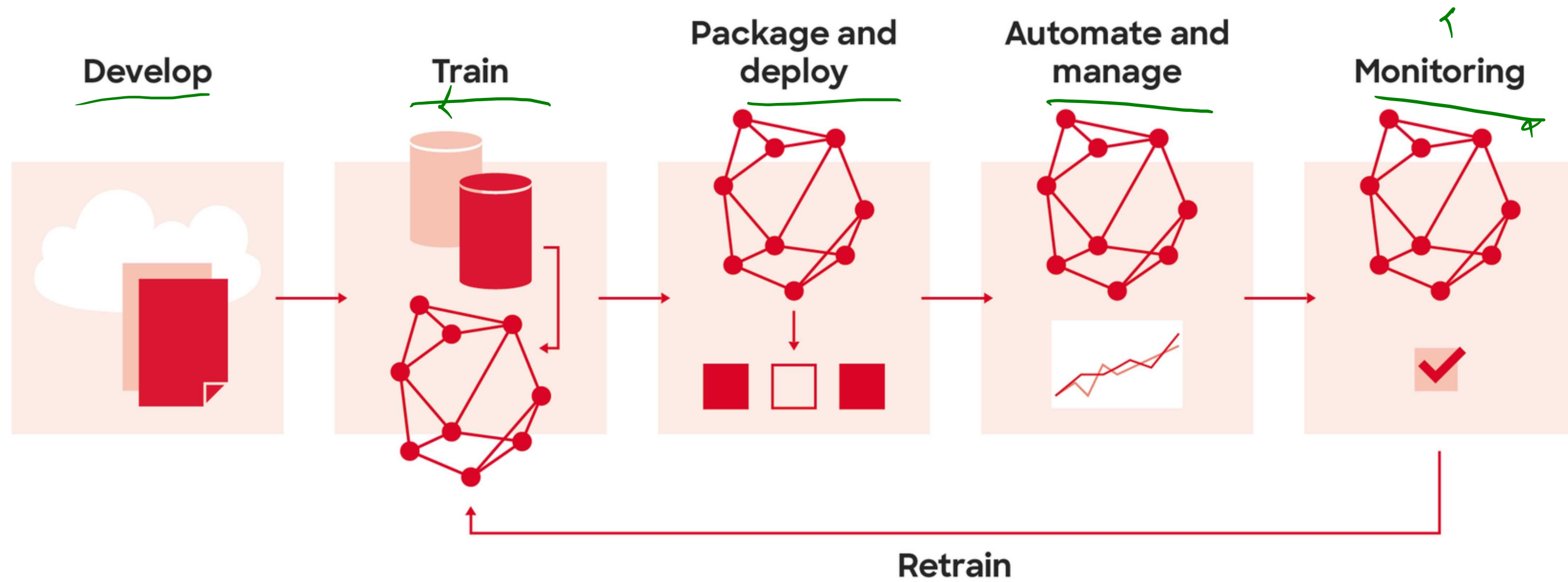
another pipeline



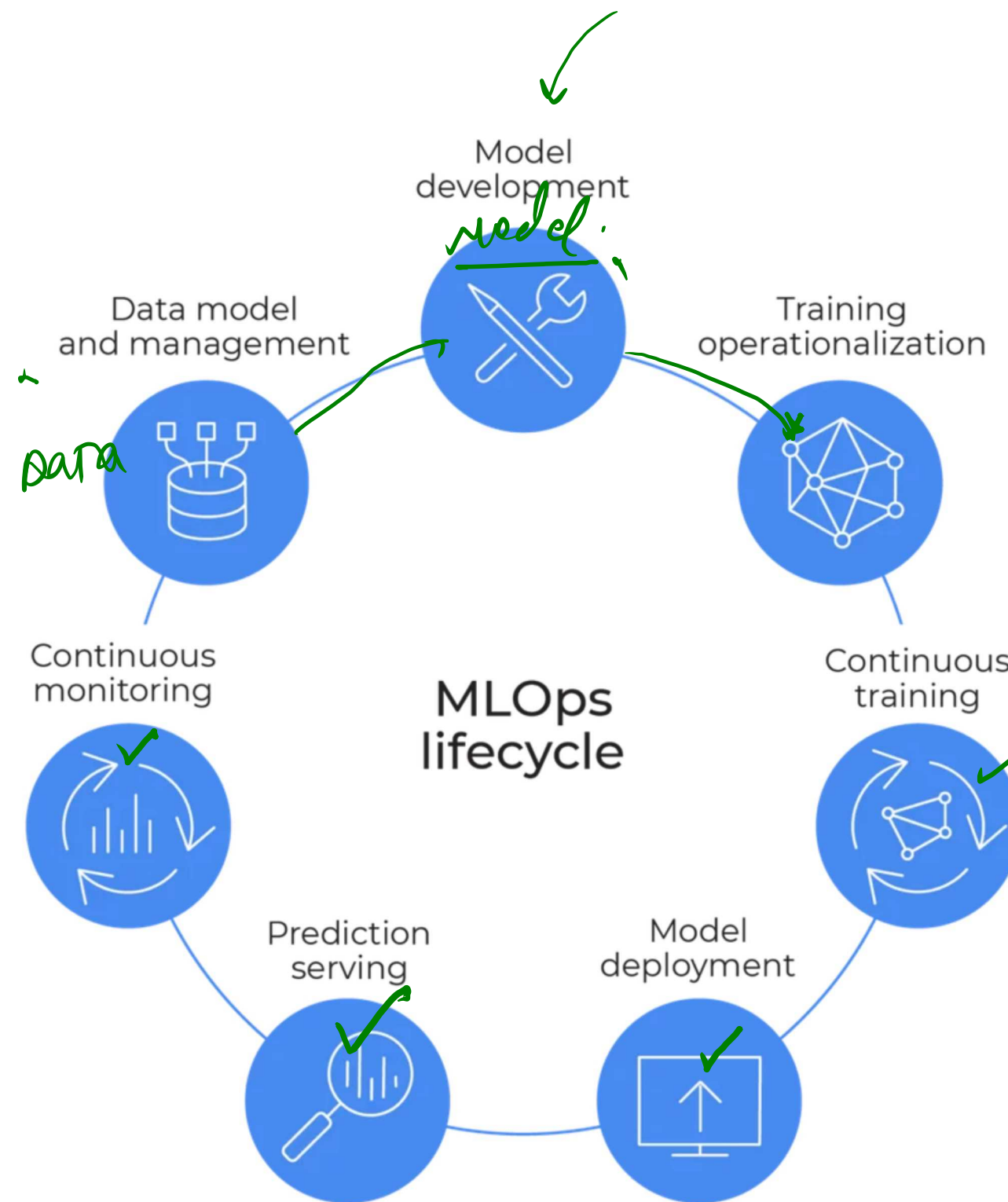




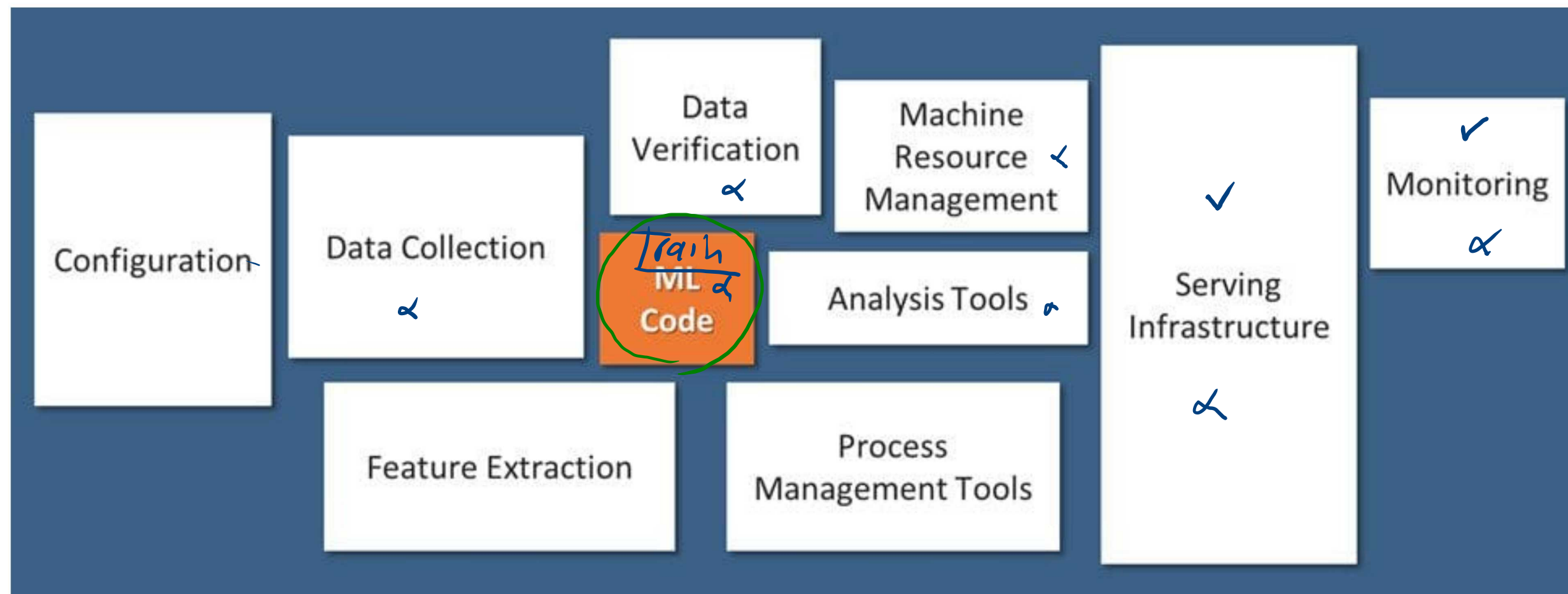
Automated ML model development life cycle



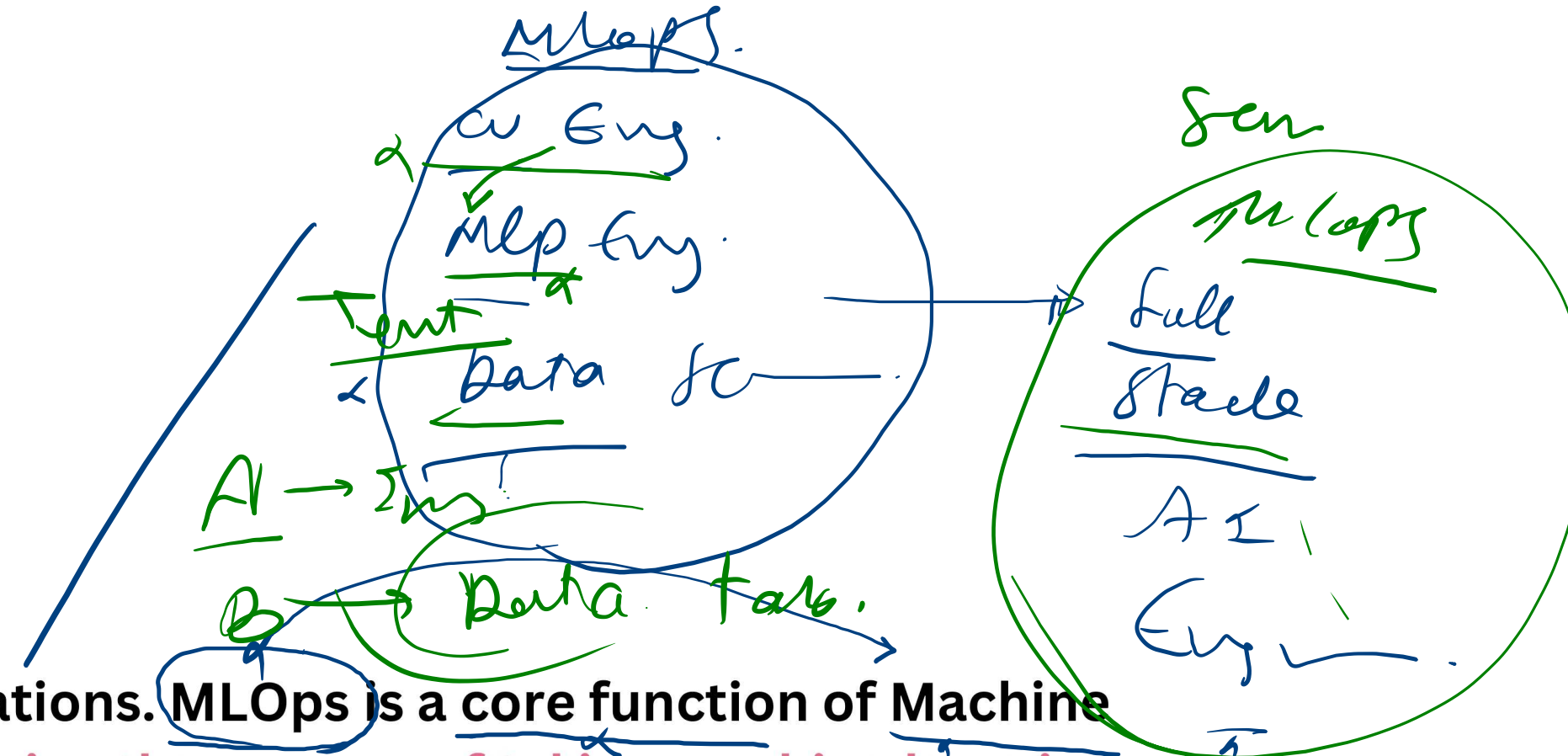
data
model Tr
eval
Deploy
monitoring



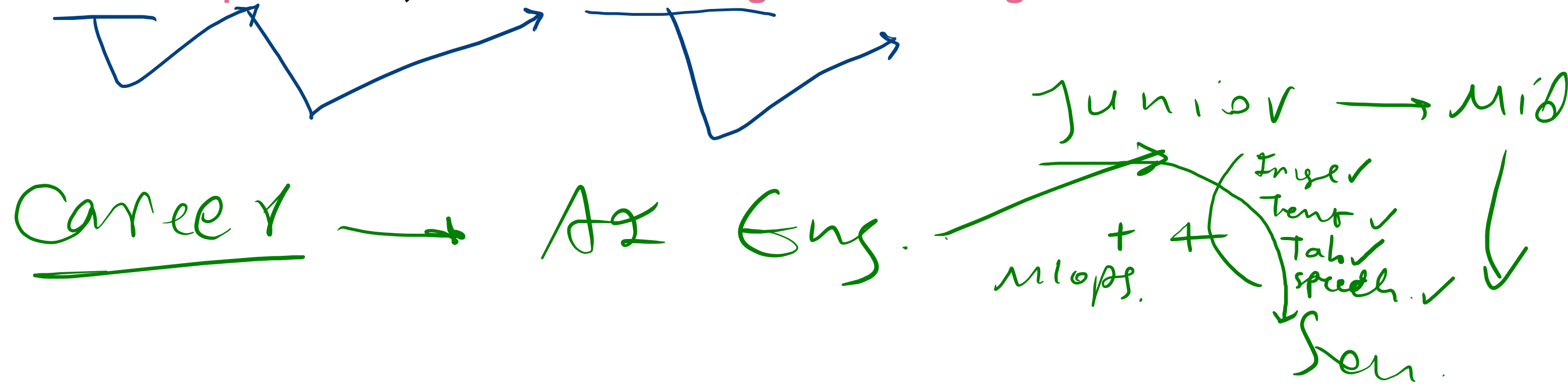
Building a ML application means implementing much more than just your ML model.



mlops + ML Eng.
 ①
 Mlops + AI Eng.

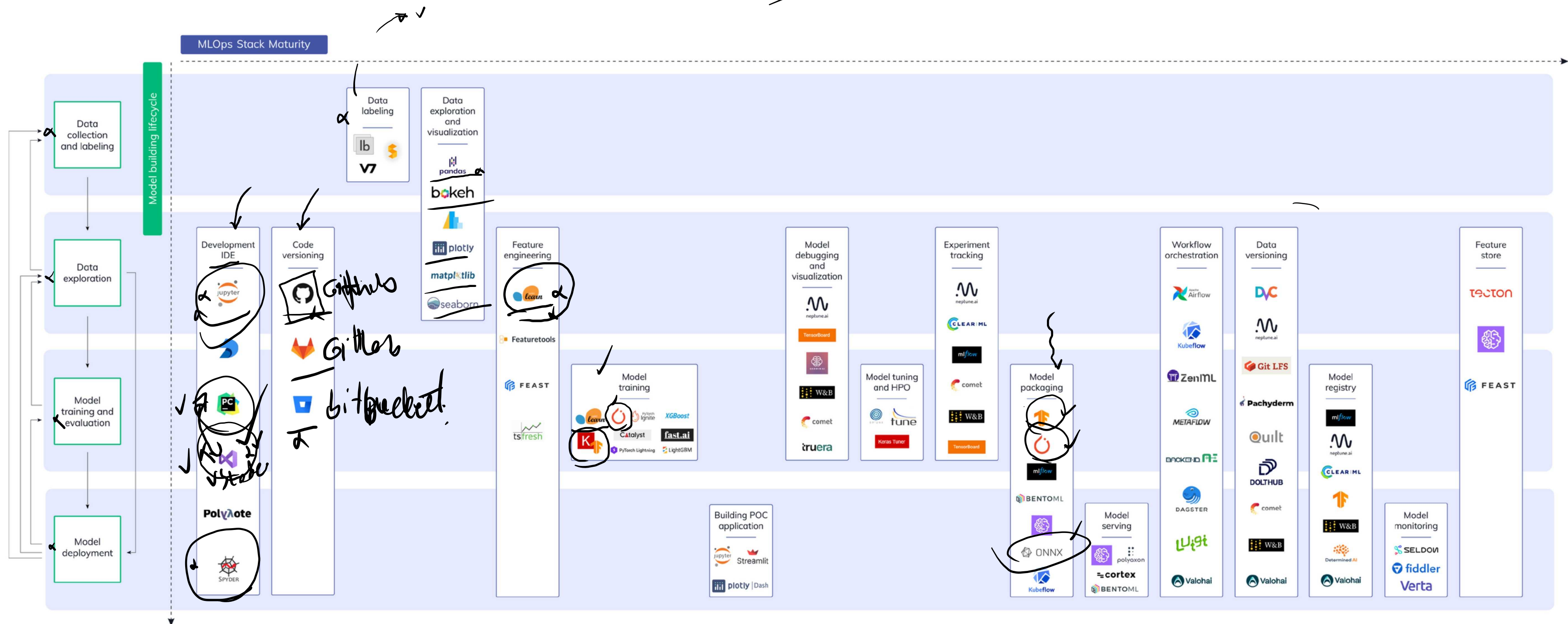


ML Ops stands for Machine Learning Operations. ML Ops is a core function of Machine Learning engineering, focused on streamlining the process of taking machine learning models to production, and then maintaining and monitoring them.

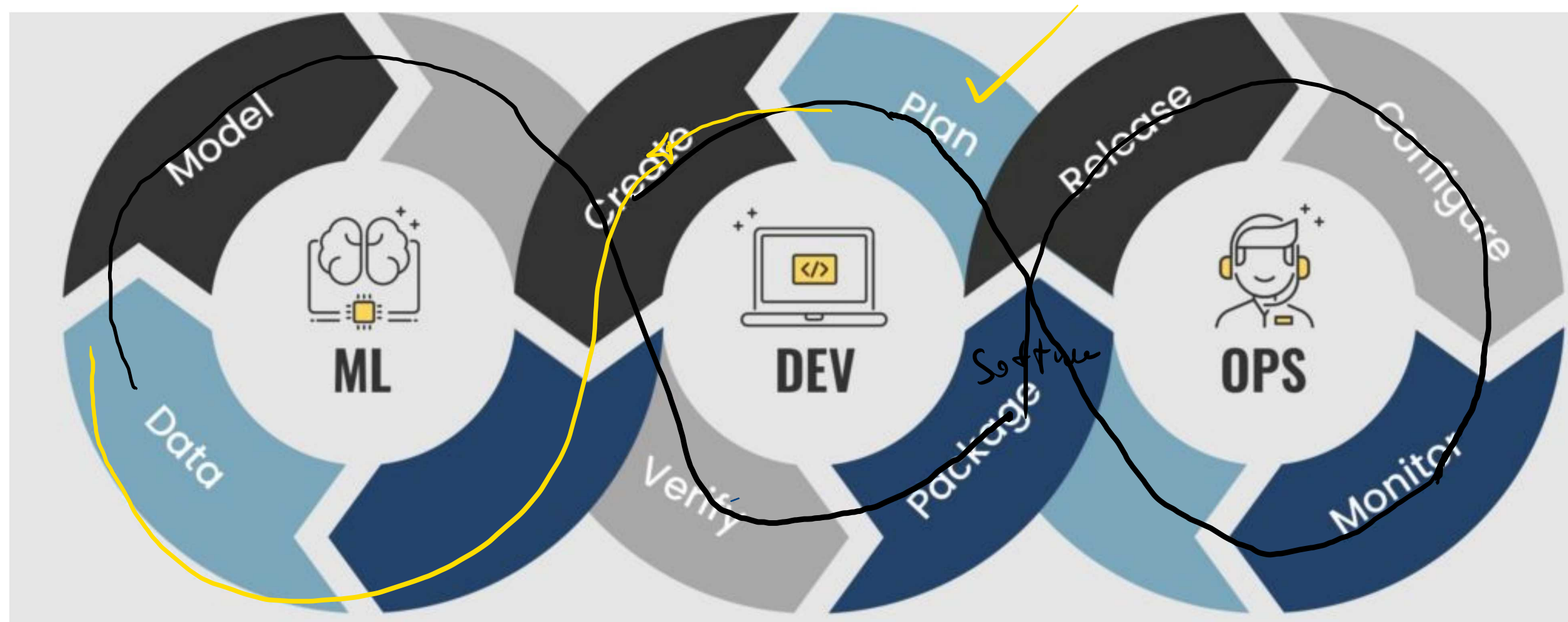


MLOps Tools

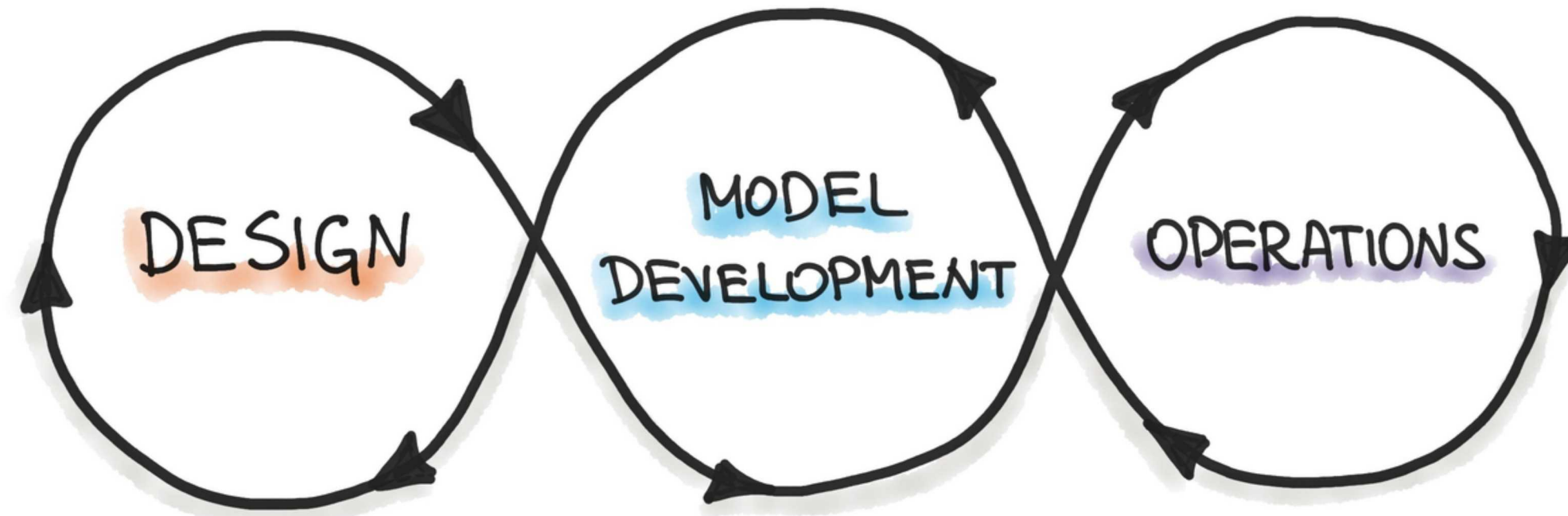
PoC
mup



oo *Dev* → *Ops* → *nonstop.*



MLOps



- Requirements Engineering
- ML Use-Cases Priorization
- Data Availability Check

- Data Engineering
- ML Model Engineering
- Model Testing & Validation

- ML Model Deployment
- CI/CD Pipelines
- Monitoring & Triggering

Machine Learning tools & platforms landscape - v.1.1 February 2021

Presented by  MLReef

DATA MANAGEMENT

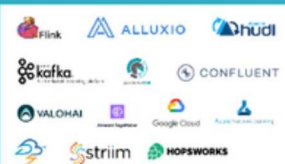
Data Exploration & Management



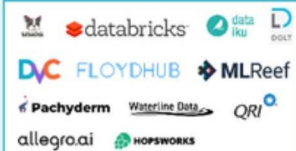
Data Labelling



Data Streaming



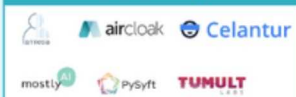
Data Version Control



Data Generation



Data Privacy



Data Quality Checks



MODELLING

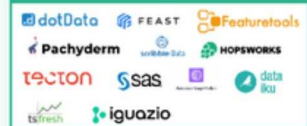
Notebook / Code Management



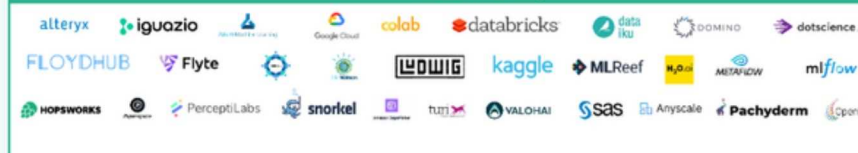
Data Processing & Visualization



Feature Engineering



Model Training



Experiment Tracking



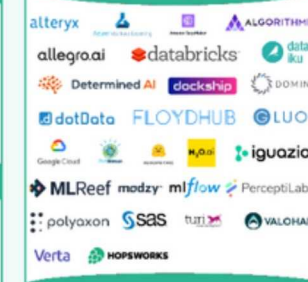
Model (Hyperparameter) Optimization



Auto ML



Model Management



Model Evaluation



Model Explainability



Frameworks & major libraries



CONTINUOUS DEPLOYMENT

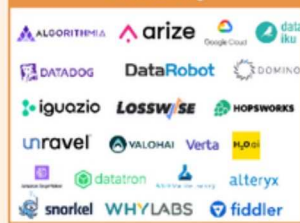
Data Flow Management



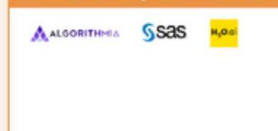
Feature Transformation



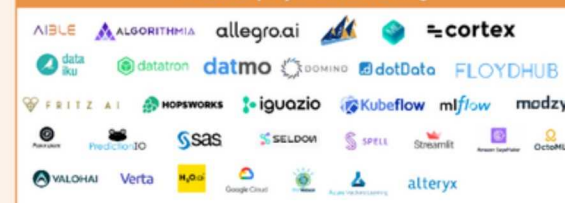
Monitoring



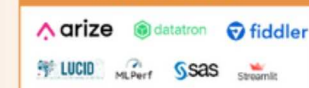
Model Compliance & Audit



Model Deployment & Serving



Model Validation



Model Compatibility



COMPUTING MANAGEMENT

Computing & Data Infrastructure



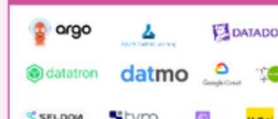
Environment Management



Resource Allocation



SCALING



SECURITY & PRIVACY





Why ?

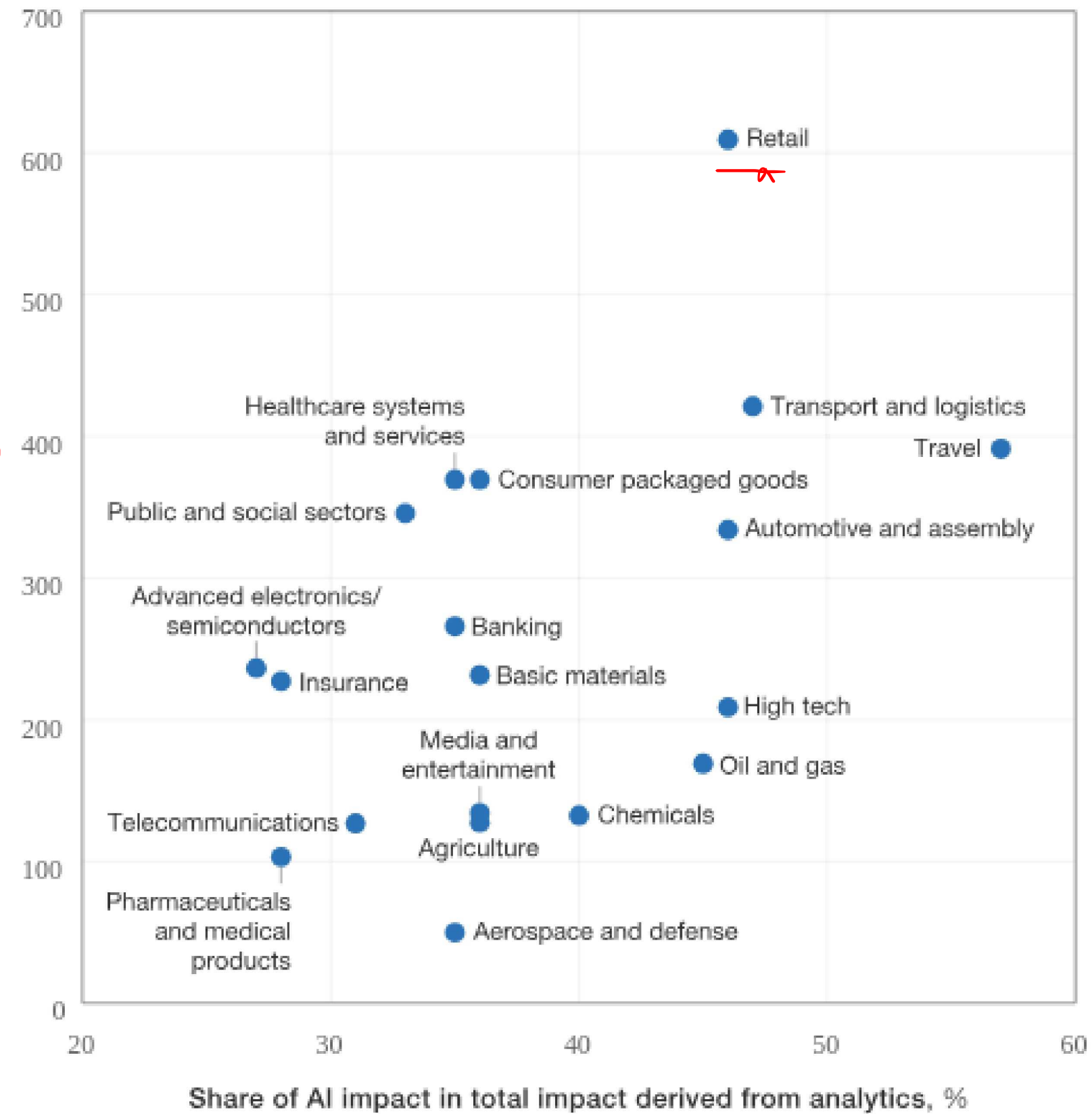
Jobs
See
Mid A In Eng.
maps.



Artificial intelligence (AI) has the potential to create value across sectors.

AEK in
Agent ✓
Value ✓

a AI impact,
\$ billion

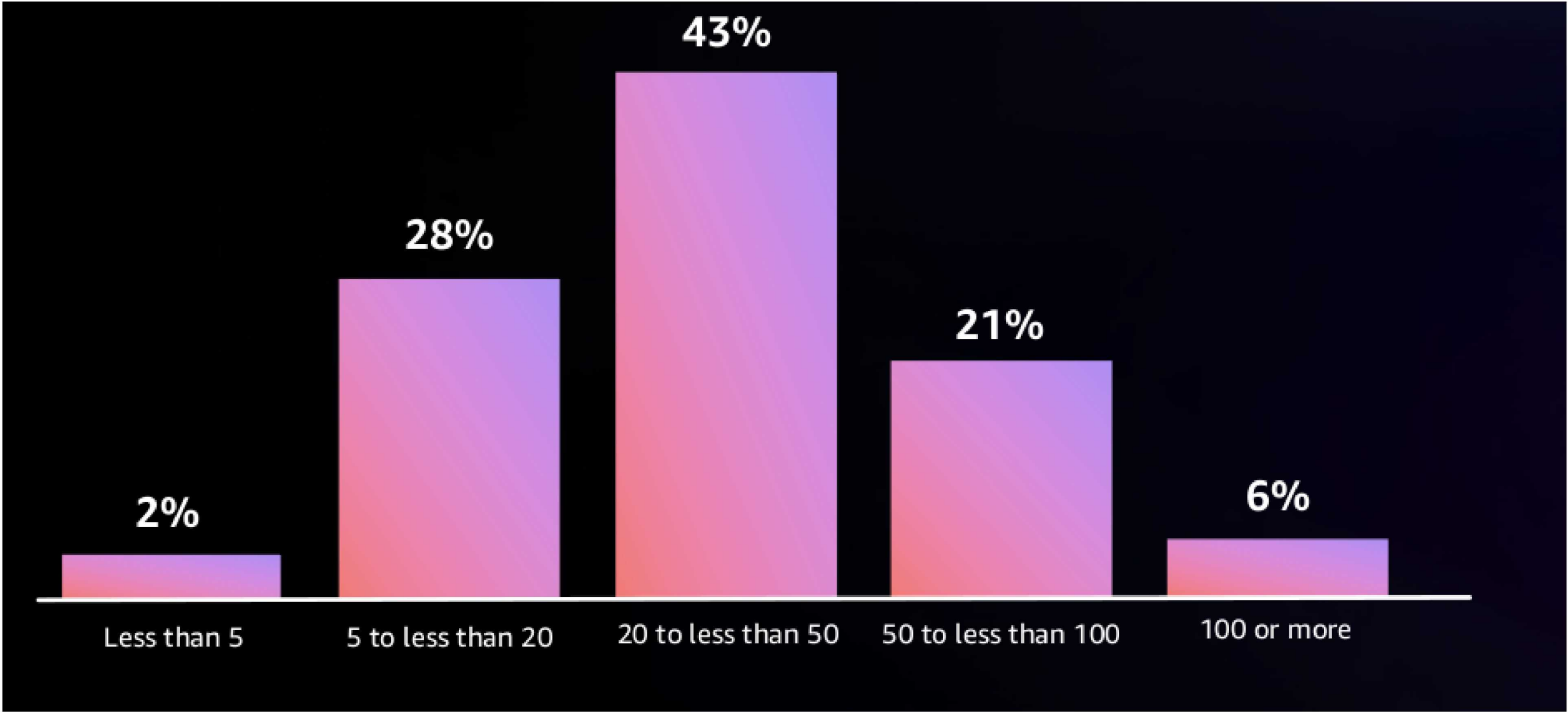


**by the end of 2024 , 75% of
enterprises will shift from
piloting to operationalizing AI**

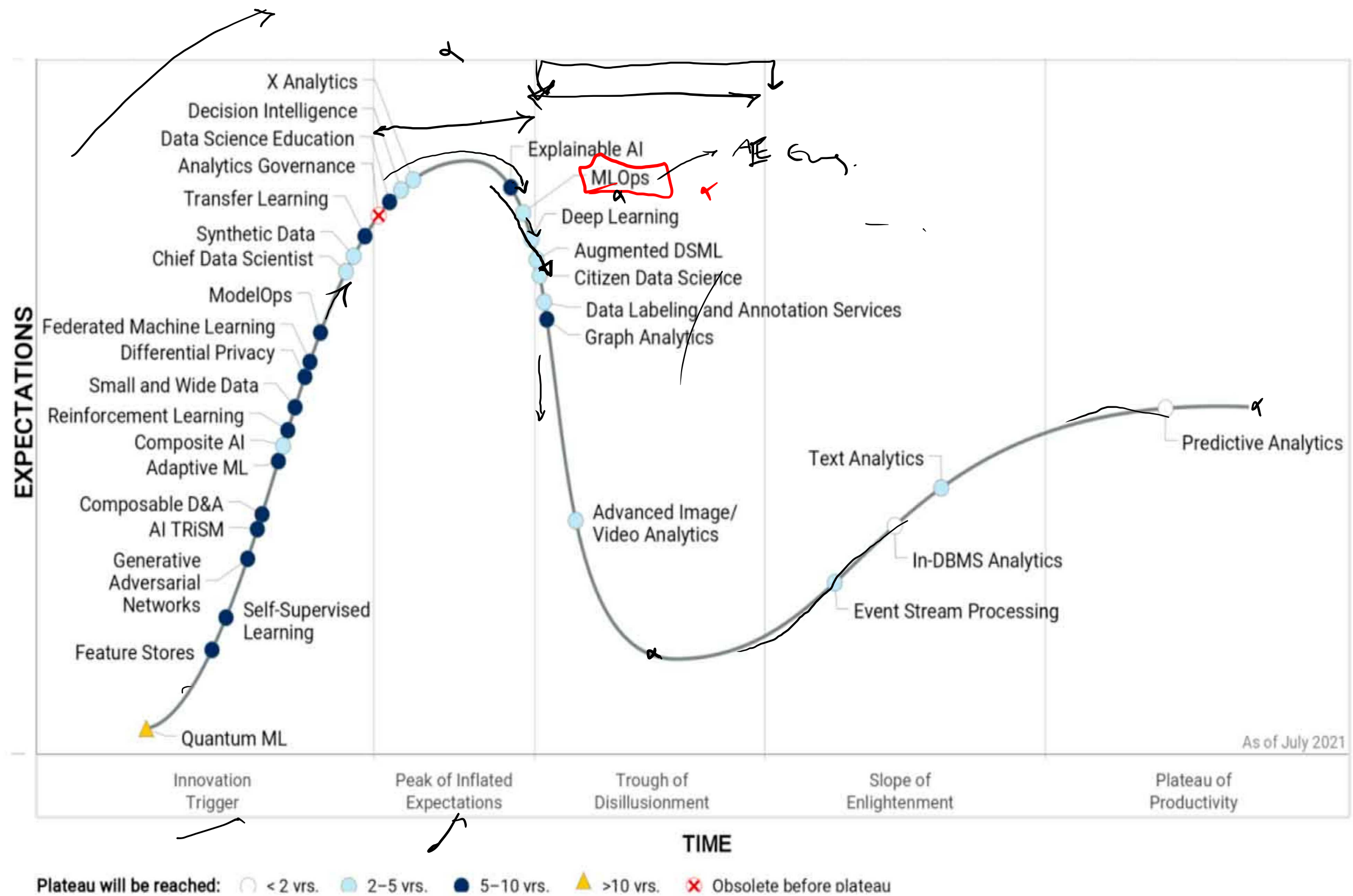
<https://www.gartner.com/en/documents/3987446>

Number of AI use cases deployed

70% OF ORGANIZATIONS (WHO USE AI) DEPLOY MORE THAN 20 AI USE CASES,
WITH GLOBAL ENTERPRISES DEPLOYING MORE CASES ON AVERAGE COMPARED
TO SMALLER SIZE ORGS

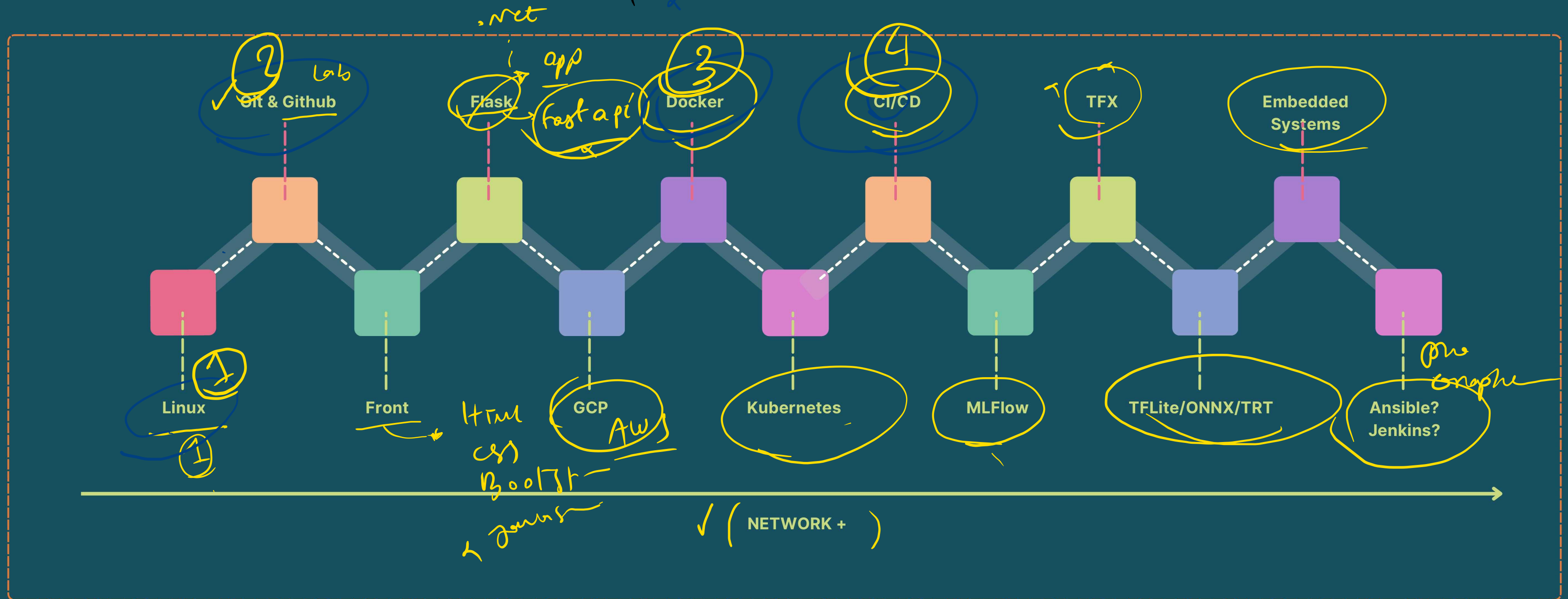


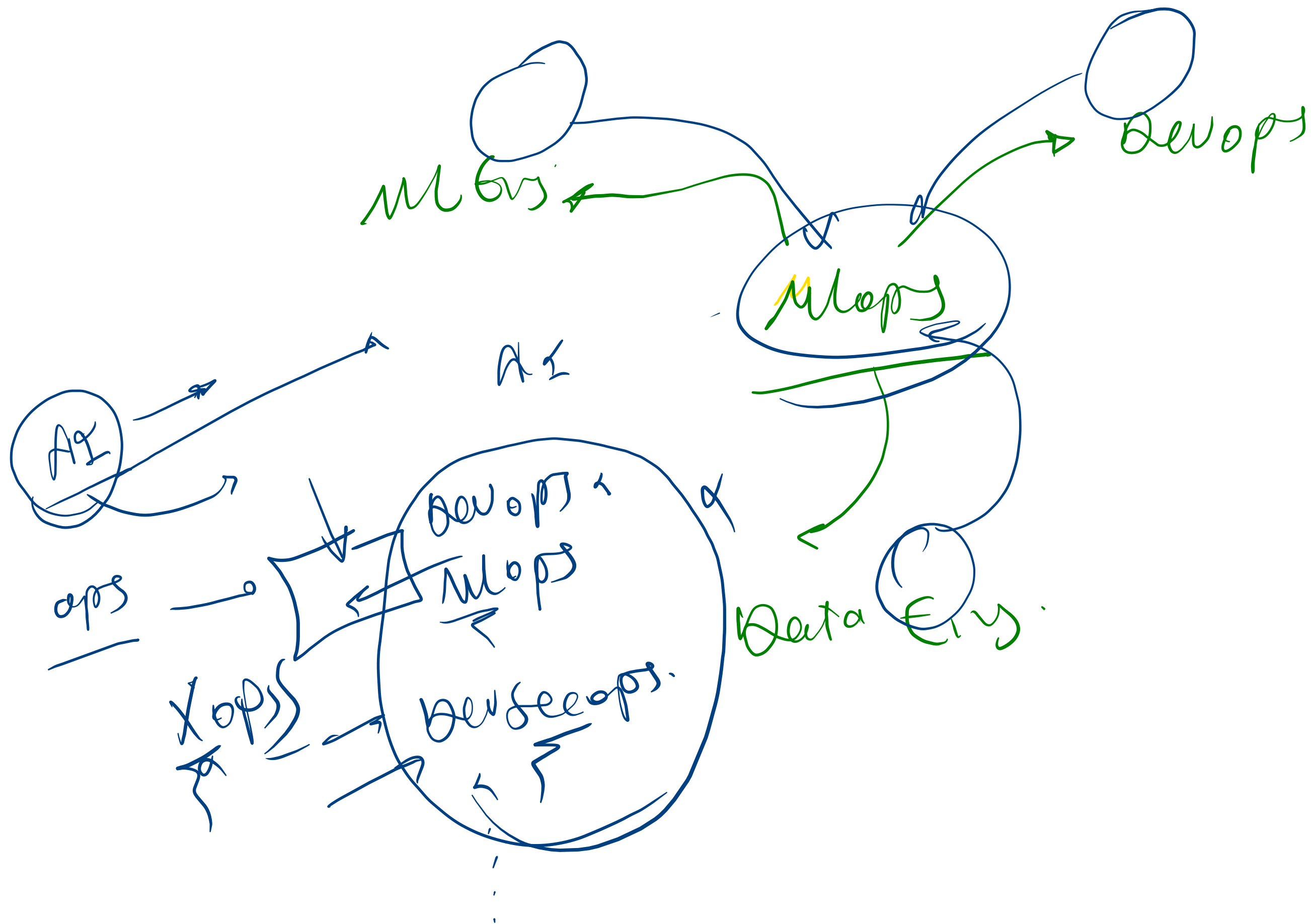
618 Companies
Source: 2022 Gartner AI Use-Case ROI Survey



How ?

Roadmap





Best Wishes